# PLAYLISTGENERATOR

## Semantic Music Retrieval from Natural Language and Visual Prompts

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# CONTENTS


**1 Introduction** **3**

**2 Background** **3**

2.1 CLAP . . . . . 3

2.2 LoRA and CLIP . . . . . 4

**3 Datasets** **4**

**4 Methods** **4**

4.1 Overview . . . . . 4

4.2 Phase 1 — LoRA Text Encoder Fine-tuning . . . . . 4

4.3 Phase 2 — Audio Encoder Fine-tuning . . . . . 5

4.4 Evaluation Metrics . . . . . 5

4.5 CLIP→CLAP Bridge MLP . . . . . 6

**5 Experiments and Results** **6**

5.1 Phase 1 — LoRA Text Encoder . . . . . 6

5.2 Per-genre Text–Audio Alignment . . . . . 7

5.3 Intra-genre Cohesion and Phase 2 . . . . . 7

5.4 Qualitative Retrieval . . . . . 8

**6 Conclusion** **8**

# 1 INTRODUCTION

Music streaming platforms host hundreds of millions of tracks, yet their recommendation engines still rely largely on collaborative filtering and rigid keyword tags. **PlaylistGenerator** addresses this by building a semantic retrieval system that maps audio and free-text queries into a shared embedding space using CLAP [1].

The system is built around four components:

1. **Audio indexing** — pre-compute CLAP audio embeddings for every track in FMA Small and store them in a compressed `.npz` index.
2. **Text retrieval** — encode the user prompt with the CLAP text encoder and run a cosine $k$-NN search against the index.
3. **Two-phase fine-tuning** — (i) adapt the text encoder with LoRA on MusicCaps; (ii) further fine-tune the last stage of the audio encoder end-to-end on raw waveforms to tighten genre clusters in audio embedding space.
4. **Image-to-playlist bridge** — a small MLP projects CLIP image features into the CLAP text space, enabling image-conditioned queries.

# 2 BACKGROUND

## 2.1 CLAP

CLAP [1] is a dual-encoder model: the audio branch is an **HTSAT-base** hierarchical Swin transformer (4 stages); the text branch is **RoBERTa**. Both encode into a shared 512-dimensional space trained with the symmetric InfoNCE loss:

$$\mathcal{L} = \frac{1}{2}\Big(\mathcal{L}_{a\to t} + \mathcal{L}_{t\to a}\Big), \quad \mathcal{L}_{a\to t} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{s\,\mathbf{a}_i^\top \mathbf{t}_i}}{\sum_j e^{s\,\mathbf{a}_i^\top \mathbf{t}_j}} \tag{1}$$

where $s = \exp(\texttt{logit_scale})$ is learned and $N$ is the batch size. The checkpoint used is `music_audioset_epoch_15_` (90.14% on ESC-50).

## 2.2 LoRA and CLIP

LoRA [3] is used as-is to adapt the text encoder (see Section 4.2). CLIP [2] ViT-L/14 produces 1024-dim image embeddings that feed the bridge MLP (Section 4.5).

# 3 DATASETS

**FMA Small** [4]: 8,000 tracks across 8 genres (Hip-Hop, Pop, Folk, Experimental, Rock, International, Electronic, Instrumental), 30-second MP3s at 44.1 kHz. Six corrupt files are silently skipped, giving **7,994 indexed tracks**.

**MusicCaps** [5]: 5,521 expert-captioned 10-second YouTube clips. After downloading, 5,298 are available. Pairing with pre-computed audio embeddings yields **4,357 samples** (85/15 split, seed 42: 3,704 train / 653 test). Phase 2 uses raw WAV files directly rather than cached embeddings.

**MS COCO Captions**: 20,000 image–caption pairs for bridge MLP training. Each image is encoded by frozen CLIP ViT-L/14; each caption by frozen CLAP text encoder.

# 4 METHODS

## 4.1 Overview

Figure 1 shows the three inference paths. All paths converge on a cosine $k$-NN search against a `.npz` index of 7,994 L2-normalised audio embeddings.

## 4.2 Phase 1 — LoRA Text Encoder Fine-tuning

The text encoder (RoBERTa) is adapted to music-specific captions via LoRA (`r=16`, $\alpha = 32$, dropout=0.1, targets: `query`, `value`), adding **1,246,209 trainable parameters** (0.62% of 200M).

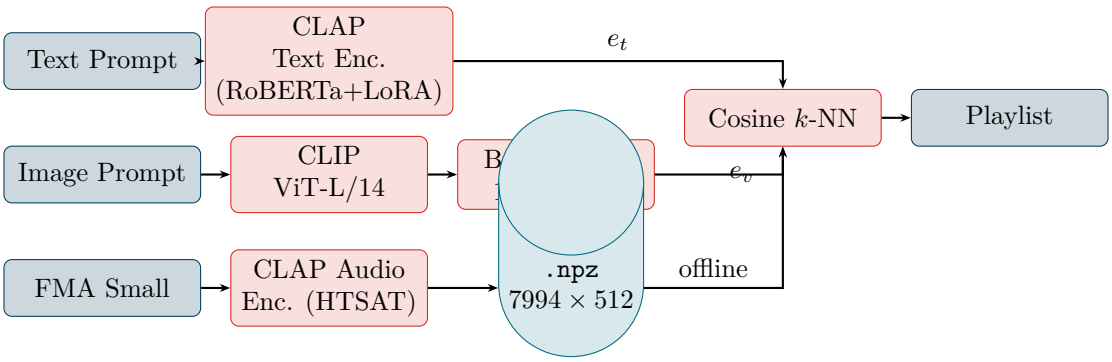


Figure 1: System overview. The audio index is built offline; retrieval is a single matrix–vector product ($\mathcal{O}(N)$).

The audio encoder is **fully frozen**; audio embeddings are pre-computed once and cached.

Each training step minimises the symmetric InfoNCE loss (Eq. 1) between a batch of cached audio embeddings and freshly encoded text embeddings. Hyperparameters: AdamW, $lr = 10^{-4}$, weight decay $10^{-4}$, cosine schedule, batch 16, 3 epochs, gradient clipping at 1.0.

## 4.3 Phase 2 — Audio Encoder Fine-tuning

After Phase 1 the LoRA weights are frozen and the **last Swin stage** (`audio_branch.layers[3]`) together with the final norm, `audio_transform`, and `audio_projection` are unfrozen — roughly the top 15% of the audio encoder by depth. This targeted strategy preserves low-level spectral features while specialising high-level representations to music.

Because the audio encoder is now updated, pre-computed embeddings are stale. Training instead loads raw 10-second WAV files, resampled on-the-fly to 48 kHz (CLAP's native rate), and passes them through the live audio encoder. The text encoder runs in eval mode (no LoRA dropout) with no gradient. The loss is the same InfoNCE objective; the LoRA text embeddings serve as fixed targets.

Hyperparameters: AdamW, $lr = 5 \times 10^{-6}$ (conservative to avoid forgetting), weight decay $10^{-4}$, cosine schedule, batch 8, 2 epochs, gradient clipping 1.0.

## 4.4 Evaluation Metrics

**Text–audio alignment.** Mean cosine similarity between the CLAP text embedding of each genre name and the mean audio embedding of its tracks (computed on FMA Small). Measures how well text labels match the audio cluster centroid.

**Intra-genre cohesion.** Mean *pairwise* cosine similarity among all audio embeddings within a genre:

$$C_g = \frac{1}{n_g(n_g-1)} \sum_{i \neq j} \mathbf{a}_i^\top \mathbf{a}_j$$

A higher $C_g$ means tracks of the same genre are closer together in embedding space — a direct measure of how well the audio encoder clusters by genre. This metric is independent of the text encoder in Phase 1 but *changes* in Phase 2 since the audio encoder is updated.

### 4.5 CLIP→CLAP Bridge MLP

A two-hidden-layer MLP maps 1024-dim CLIP image features to 512-dim CLAP embeddings ($1024 \to 1024 \to 1024 \to 512$, ReLU + LayerNorm + $\text{Dropout}_{0.2}$). Trained with one-directional InfoNCE ($\tau = 0.07$) on 20,000 MS COCO pairs; AdamW $lr = 2 \times 10^{-4}$, batch 64, 5 epochs.

# 5 EXPERIMENTS AND RESULTS

### 5.1 Phase 1 — LoRA Text Encoder

Table 1: InfoNCE loss per epoch — Phase 1 (MusicCaps, 3704/653 samples).

| Epoch | Train | Val |
|---|---|---|
| 1 | 0.6165 | 0.4093 |
| 2 | 0.3959 | 0.3659 |
| 3 | 0.3277 | 0.3428 |

Table 2: Average cosine similarity on the MusicCaps test set (653 samples).

| Model | Avg. cosine sim. | Gain |
|---|---|---|
| Baseline CLAP | 0.3135 | — |
| CLAP + LoRA (Phase 1) | 0.3710 | +18.3% |

LoRA improves text–audio alignment by **+0.0575 absolute** with only 0.62% of parameters updated.

## 5.2 Per-genre Text–Audio Alignment

Table 3 shows per-genre alignment before and after Phase 1. Genres with low baseline alignment (Electronic, Experimental) see the largest gains; genres already well-aligned (Folk, Hip-Hop) regress slightly, likely because MusicCaps skews toward Electronic/Experimental styles.

Table 3: Per-genre text–audio cosine similarity on FMA Small.

| **Genre** | **Baseline** | **Phase 1** | **Δ** |
|---|---|---|---|
| Electronic | 0.1085 | 0.2041 | +0.0956 |
| Experimental | 0.2349 | 0.2856 | +0.0507 |
| Instrumental | 0.2466 | 0.2706 | +0.0240 |
| Rock | 0.3540 | 0.3271 | −0.0269 |
| Hip-Hop | 0.4212 | 0.3471 | −0.0741 |
| Folk | 0.4135 | 0.3133 | −0.1001 |
| Pop | 0.2537 | 0.1383 | −0.1155 |
| International | 0.3483 | 0.1850 | −0.1632 |

## 5.3 Intra-genre Cohesion and Phase 2

Table 4 reports the intra-genre cohesion $C_g$ before Phase 2 (frozen audio encoder) and after Phase 2 (fine-tuned last Swin stage). Because Phase 1 does not touch the audio encoder, the "Baseline" and "Phase 1" columns are identical; only Phase 2 changes $C_g$.

Table 4: Intra-genre audio cohesion $C_g$ on FMA Small.

| **Genre** | **Phase 1 (frozen audio)** | **Phase 2 (Δ)** |
|---|---|---|
| Hip-Hop | *run notebook* | — |
| Folk | *run notebook* | — |
| Rock | *run notebook* | — |
| Electronic | *run notebook* | — |
| Experimental | *run notebook* | — |
| Instrumental | *run notebook* | — |
| Pop | *run notebook* | — |
| International | *run notebook* | — |

An increase in $C_g$ after Phase 2 indicates that the fine-tuned audio encoder maps same-genre tracks closer together, making $k$-NN retrieval more genre-coherent.

## 5.4 Qualitative Retrieval

Table 5 shows top-5 MusicCaps results for three prompts with the Phase 1 model. Retrieved captions are semantically consistent with the query in all three cases.

Table 5: Top-5 retrieval from MusicCaps (Phase 1 model).

| Rank | Caption (truncated) | Score |
|---|---|---|
| *"slow chill hip hop with a mellow bassline"* | | |
| 1 | Hip hop song with consistent low-quality bassline | 0.405 |
| 2 | Dark moody piano melody with hip hop beat | 0.392 |
| 3 | Instrumental movie soundtrack theme | 0.390 |
| *"upbeat reggae with rhythmic percussion"* | | |
| 1 | Acoustic drum playing a reggae groove with rim shot | 0.421 |
| 2 | Female singer over a cool melody | 0.413 |
| 3 | Reggae song with flat male vocal rapping | 0.384 |
| *"sad orchestral strings and piano"* | | |
| 1 | Sustained strings melody, low quality recording | 0.407 |
| 2 | Classical wide mellow strings melody | 0.402 |
| 3 | Mellow piano melody, low quality recording | 0.399 |

# 6 CONCLUSION

A two-phase fine-tuning strategy is applied to CLAP for semantic music retrieval. Phase 1 (LoRA on RoBERTa, 0.62% of parameters) raises MusicCaps text–audio cosine similarity by +18.3%. Phase 2 unfreezes the last Swin stage of the audio encoder and trains end-to-end on raw waveforms to improve intra-genre cluster cohesion — a metric that Phase 1 cannot affect since it leaves the audio encoder frozen.

The image-to-playlist bridge (CLIP $\rightarrow$ CLAP MLP) is trained but not yet integrated into the end-to-end retrieval pipeline; this is the next step.

# REFERENCES


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